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10 October 1963
Superseding
MIL-A-9597A(USAF)
22 November 1960

MILITARY SPECIFICATION

ANTENNA AT-309( )/GPX

1. SCOPE

1.1 This specification covers one type of antenna designated Antenna AT-309( )/GPX, hereinafter referred to as "the antenna".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or requests for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4158	Electronic Equipment, Ground; General Requirements For
MIL-T-4807	Tests; Vibration and Shock Ground Electronic Equipment (Requirements For)
MIL-N-7513	Nomenclature Assignment and Nameplate Approval; Contractors Method For Obtaining
MIL-T-9107	Test Reports; Preparation Of
MIL-F-14072	Finishes For Ground Signal Equipment

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-810	Environmental Test Methods For Aerospace and Ground Equipments

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC-5985

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3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 General specification.- The requirements of MIL-E-4158 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.2.1 Exceptions to the requirements of MIL-E-4158

3.2.1.1 Service conditions.- The antenna shall be designed and constructed to operate under any combination of the service conditions specified in the general specification with exceptions and additions as follows.

3.2.1.1.1 Climatic

a. Ambient temperature: The antenna shall be designed and constructed to operate in a "Cold Weather Area" and in "Desert and Tropical Areas."

b. The antenna shall be designed and constructed to meet the minimum performance requirements during, and after exposure to the following environmental conditions, as detailed in section 4 of this specification:

(1) Salt fog
(2) Fungus resistance
(3) Rain
(4) Sand and dust

c. Winds

(1) Operating (rotating): 52 knots no ice

(2) Nonoperating (non-rotating): 87 knots no ice

d. Ice load: Two inches of glaze ice measured radially to all surfaces without wind.

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3.2.2 Lightweight material.- Wherever practical, lightweight high-strength materials shall be used.

3.2.3 Qualification of welders.- The requirement for qualification of welders shall not apply.

3.2.4 Service life.- Theantenna shall be capable of operating 23 hours per day for a period of 2000 hours with only normal maintenance and without major overhaul. The operational period may start after an initial storage period not exceeding 2 years.

3.3 Nomenclature.- Nomenclature shall be obtained in accordance with the requirements of MIL-N-7513. The parentheses appearing in the nomenclature will be dropped, or replaced by a letter to designate the particular design.

3.4 Design.- The antenna shall be designed and constructed of lightweight high strength materials, with the antenna itself designed to act as a structural beam.

3.4.1 Design data.- The contractor shall provide to the procuring activity one set of reproducible manufacturing drawings suitable for fabrication and construction of this antenna.

3.4.2 Base of assembly.- The design and construction of the antenna shall be such as to facilitate field assembly without the use of special tools. The antenna design shall lend itself to ease of assembly and disassembly. When disassembled, the antenna array shall consist of 4 sections of equal size. Particular attention shall be given in the design of the antenna so that it can be assembled and installed by untrained personnel under field conditions. The antenna design shall be such that it will withstand rough handling.

3.4.3 Loading.- In addition to its own weight, the antenna shall be designed to withstand the following loads:

a. Wind velocity of 52 knots, with no ice, while rotating about the center of gravity at 10 revolutions per minute (RPM).

b. Wind velocity of 87 knots, with no ice, supported at designated support points, (See 6.3) nonrotating.

c. Two inches of ice measured radially to surfaces without wind.

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Any deflection of the antenna resulting from the above loads shall be such that it will not detract from the performance requirements specified herein.

3.4.4 Mounting provisions.- Information concerning the design of mounting provisions for the antenna will be provided by the procuring activity. (See 6.3.)

3.5 Performance.- The antenna shall be an open box array. The antenna shall meet the following requirements over the frequency band of 1000 to 1120 megacycles (MC) unless otherwise specified.

3.5.1 Polarization.- The antenna shall be vertically polarized.

3.5.2 Voltage standing wave ratio (VSWR).- The VSWR shall not exceed 1.5:1 at the frequencies of 1020 MC through 1040 MC and 1080 MC through 1100 MC. The VSWR at all other frequencies in the band from 1000 MC to 1120 MC shall not exceed 2:1. This requirement shall be true when the antenna is fed from a coaxial line of 52 ohms characteristic impedance.

3.5.3 Horizontal pattern.- The horizontal pattern shall have a main lobe beam width of 2.4 degrees or less at the 3 DB below maximum points and shall be less than 7.5 degrees at the 20 DB below maximum points at frequencies of 1000 MC to 1120 MC inclusive. The side and back lobes shall be:*

a. 26 DB or more below the main beam maximum at 1030 MC and 1090 MC.

b. 25 DB or more below the main beam maximum at 1020 MC, 1040 MC, 1080 MC and 1100 MC.

c. 23 DB or more below the main beam maximum at all other frequencies in the band from 1000 MC to 1120 MC

*Note: Up to an elevation of 30 degrees above the horizontal and as low as possible at higher elevation angles.

3.5.4 Vertical pattern.- The vertical pattern shall have a symmetrical main lobe with a width of 45 degrees plus or minus 5 degrees as measured at the 3 DB down from maximum points. The main lobe pattern shall be no more than 4 DB below maximum at the elevation of 40 degrees above the

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horizontal and no more than 15 DB below maximum at an elevation of 60 degrees above the horizontal. The main lobe shall decrease symmetrically from plus 60 degrees elevation to where any radiated energy at elevation angles of plus 90 degrees to 150 degrees shall be at least 22 DB below main lobe maximum. The radiated energy at elevation angles of plus 150 degrees to 180 degrees shall be at least 30 DB below the main lobe maximum. The radiated energy at elevation angles of minus 90 degrees and greater shall be at least 30 DB below the main lobe maximum.

3.5.5 Power output.- The antenna shall be capable of radiating without breakdown, radio-frequency (R-F) pulses of 30 kilowatts (KW) peak power at an average duty cycle of 0.7 percent.

3.5.6 Input filter.- The antenna shall have a filter in the input providing for at least 50 DB attenuation to signals at all frequencies in the band from 1280 MC to 11,000 MC inclusive and shall provide no more than 0.5 DB attenuation to frequencies in the 1000 MC to 1120 MC band inclusive.

3.5.7 Input connector.- The input connector shall be a female "NH" type connector. A cable clamp shall be provided to relieve any strain on the connector.

3.5.8 Antenna feeders.- Strip line printed circuit technique shall be utilized for the antenna feed system. The antenna feeder is defined as the transmission line system which will provide the division of power to the radiating elements.

3.5.9 Gain.- The gain of the antenna shall be at least 22 DB above that of an isotropic.

3.6 Dimensions.- The antenna shall be a nominal 28 feet long, 18 inches high and 20 inches deep. It shall be divided into four sub-sections with suitable mechanical connections to assure sufficient structural rigidity to be capable of fulfilling the requirements of this specification when supported by a three point mounting bracket.

3.7 Weight.- The weight of the antenna shall not exceed 325 pounds.

3.8 Color and finish.- The antenna shall be finished in accordance with MIL-F-14072. The final paint coat shall be lusterless green, color No. 34087 in accordance with FED-STD-595.

3.9 Interchangeability of parts.- All individual parts shall be directly

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interchangeable for like parts between antennas or from stock. Subsections referenced in 3.6 shall be mechanically interchangeable between like sections and shall cause no more than 1 DB deterioration in the new antenna assembly electrical performance as specified in 3.5 and its sub paragraphs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. (See 6.2.d.)

4.2 Classification of tests.- The inspection and testing of antenna shall be classified as follows:

a. Preproduction tests.....(See 4.4)
b. Acceptance tests.....(See 4.5)

4.3 Test conditions.- Unless otherwise specified, all tests required by this specification shall be made at prevailing ambient temperature and humidity conditions.

4.4 Preproduction testing

4.4.1 Preproduction test samples - The preproduction test samples shall consist of models representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Preproduction test reports.- When the contractor conducts the preproduction tests, he shall, after completion of the tests, prepare a preproduction test report according to MIL-T-9107, and furnish three complete copies of the report to the procuring activity.

4.4.3 Preproduction tests.- The preproduction tests shall consist of all tests described under 4.6 "Test methods".

4.5 Acceptance tests.- Each antenna shall be subjected to the acceptance tests. The acceptance tests shall consist of the following tests as described under 4.6 "Test methods" of this specification:

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a. Mechanical inspection tests.....(See 4.6.1)
b. Electrical tests.....(See 4.6.2)
c. Performance tests.....(See 4.6.3)

4.6 Test methods

4.6.1 Mechanical inspection tests.- The antenna shall be given a thorough mechanical and visual inspection and test to determine that the quality of all materials and workmanship are in compliance with the requirements of this specification. Particular attention shall be given to the following:

a. Completeness
b. Nameplates, identification markings and labels
c. Ease of operation of gears, adjustable and sliding parts, thumb screws, controls,and switches
d. Finishes
e. Welded joints
f. Check of solder joints
g. The fit of components in their respective positions
h. Check of mounting means
i. Check of lubrication and rust prevention
j. Check of safety features and interlocks
k. Loose fastening and securing devices or parts
l. Accessibility of components and parts for servicing
m. Cable runs between components including plugs and receptacles
n. Grounding connections
o. Over-all dimensions check
p. Weight check
q. Other visual defects.

4.6.2 Electrical tests.- The antenna shall be given all electrical tests necessary to confirm that all circuits are inherently sound and in compliance with the requirements of this specification. The electric - al tests shall include, but not necessarily be restricted to, the follow- ing.

4.6.2.1 Continuity.- Each electrical component and each cord and cable shall be given a continuity test to ascertain if it is wired and connected correctly and that good electrical contact is obtained.

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4.6.2.2 Operating voltages.- The operating voltage at all important points shall be checked for conformance to those shown on the circuit label and schematic drawing. This shall be done with all controls set for normal operation of the equipment.

4.6.3 Performance tests.- The antenna shall be given all performance tests necessary to confirm that the equipment meets the performance requirements specified in 3.5.

4.6.4 Environmental tests.- The antenna shall be subjected to the following environmental conditions in accordance with MIL-STD-810, with exceptions as follows:

a. High temperature test- Method 501: Except maintain the antenna at a temperature of plus 71°C for 48 hours reduce the temperature to plus 54°C for 24 hours, then operate the antenna and record results.

b. Low temperature test- Method 502: Except maintain the antenna at a temperature of minus 62°C for 48 hours, raise the temperature to minus 54°C for 24 hours, then operate the antenna and record results.

c. Humidity test. Method 507

d. Low pressure test: Method 500 except maintain the antenna at an internal absolute pressure of 5.54 inches of mercury for 4 hours, raise the pressure to 16.8 inches of mercury for 4 hours, then operate the antenna and record results.

e. Salt fog test- Method 509: Except duration of test shall be 100 hours.

f. Fungus resistance test. Method 508

g. Rain test. Method 506

h. Sand and dust test. Method 510

i. All weather capability test- The purpose of the test is to prove the operational capability of the antenna in weather conditions involving combined fluctuations of barometric pressure, temperature and humidity. The antenna assembly shall be submitted to the following test, totally assembled in its normal operational mode.

(1) An antenna pattern at 1030 MC shall be recorded prior to performance of the remainder of this test.

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(2) The antenna shall immediately be subjected to 10 cycles of the following environmental conditions, one cycle being comprized as follows: Two hours at 20°F, barometric pressure of 31 inches of mercury and uncontrolled humidity followed immediately by two hours at 125°F, barometric pressure of 28.0 inches of mercury and 97 degrees humidity. One cycle shall be considered completed at the end of the heat whereby the beginning of the next cycle shall immediately begin with the required low temperature (20°F, et cetera) and following the above out-lined sequence until the required 10 cycles are completed. Following completion of the 10th cycle, an antenna pattern at 1030 MC shall be immediately recorded without antenna movement or change. The applicable electrical requirements of 3.5 and its subparagraphs shall be met.

(3) A substitute test program which has the approval of the procuring activity may be performed in lieu of the above program. (See 6.2.c.)

4.6.5 Mechanical tests.- The antenna shall be subjected to the following mechanical tests in accordance with MIL-T-4807:

a. Vibration test: Method 1
b. Bounce test: Method 2.

4.6.6 Wind load test.- Static loads shall be simulated on the antenna as follows:

a. Nonrotating: 87 knots
b. Rotating: (10 RPM) in a 52 knot wind

Deflection shall not cause the electrical performance of the antenna to be below that specified in 3.5. After application of the nonrotating wind loads the antenna shall not have suffered any failure or permanent deformation.

4.6.7 Life test.- The antenna shall be operated continuously for a period of 500 hours under loading, voltage, and current conditions corresponding to actual field conditions. Any item which shows defects during the test shall be perfected or replaced by a part free from such defects. Should the antenna or any of its components fail to pass the tests, the tests shall be terminated until such defects have been corrected.

5. PREPARATION FOR DELIVERY

5.1 The antenna shall be preserved, packaged and packed in accordance with the instructions of the procuring activity.

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6. NOTES

6.1 Intended use.- Antenna AT-309( )/GPX is intended for use as a high resolution transmitting and receiving antenna for identification friend or foe (IFF) equipment.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Level of preservation, packaging and packing required. (See 5.)

c. Alternate test program (See 4.6.4 i)

d. Whether the testing will be conducted by the contractor or the procuring activity (See 4.1)

6.3 Additional information required for design characteristics as specified in 3.4.4 will be furnished or made available to the contractor upon his specific request to the procuring activity.

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